

Lecture-5

I. Geometry of Periodic signals.

Recall: $(x, y) = \int_0^T x(t) y^*(t) dt$

$$\|x\| = \sqrt{(x, x)} \geq 0$$

$$E = \|x\|^2$$

$$\|ax\|^2 = \int_0^T |ax(t)|^2 dt = |a|^2 \|x\|^2.$$

We say that x, y are orthogonal if $(x, y) = 0$ (i.e. $(y, x) = 0$)

We say $x_1, x_2, \dots, x_N$ (all are T -periodic) forms an orthonormal set if:

$$\|x_i\|^2 = 1 \quad \forall i$$

$$(x_i, x_j) = 0 \quad \forall i \neq j.$$

$$\left. \begin{array}{l} \|x_i\|^2 = 1 \quad \forall i \\ (x_i, x_j) = 0 \quad \forall i \neq j. \end{array} \right\} (x_i, x_j) = \delta_{ij}$$

$$\left. \begin{array}{l} \delta_{ij} : \text{Kronecker delta} \\ \delta_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases} \end{array} \right\}$$

Example: For $n = 0, \pm 1, \pm 2, \dots, \pm N$

$$x_n = \frac{1}{\sqrt{T}} e^{j\omega n t}, \quad \omega = \frac{2\pi}{T}, \quad T = \frac{2\pi}{\omega}$$

→ orthonormal set.

If $n=m$:

$$\begin{aligned} (x_n, x_m) &= (x_n, x_n) \\ &= \int_0^T |x_n(t)|^2 dt \\ &= \int_0^T \frac{1}{T} dt = 1 \end{aligned}$$

If $n \neq m$:

$$\begin{aligned} (x_n, x_m) &= \int_0^T \frac{1}{\sqrt{T}} e^{j\omega n t} \cdot \frac{1}{\sqrt{T}} e^{-j\omega m t} dt \\ &= \int_0^T \frac{1}{T} e^{j\omega (n-m)t} dt \end{aligned}$$

$$= \frac{1}{T} \int_0^T \frac{e^{j\omega(n-m)t}}{j\omega(n-m)} dt$$

$$= 0$$

Claim: x, y are T -periodic and $(x, y) = 0$.

$\Downarrow$

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2$$

Proof: $\|x + y\|^2 = (x + y, x + y)$

$$= (x, x) + (y, y) + \underbrace{(y, x)}_0 + \underbrace{(x, y)}_0$$

$$= \|x\|^2 + \|y\|^2.$$

In general: if $x_1, \dots, x_N$ are pairwise orthogonal

$$\|x_1 + \dots + x_N\|^2 = \|x_1\|^2 + \dots + \|x_N\|^2.$$

Corollary: Suppose $x = \sum_{n=-N}^{+N} c_n \cdot e^{j\omega n t}$, $T = \frac{2\pi}{\omega}$

$$x(t) = \sum_{n=-N}^{+N} c_n \cdot \sqrt{T} \cdot \left(\frac{1}{\sqrt{T}} \cdot e^{j\omega n t} \right)$$

Note that: for $n \neq m$

$c_n \sqrt{T} \cdot \frac{1}{\sqrt{T}} e^{j\omega n t}$, $c_m \sqrt{T} \cdot \frac{1}{\sqrt{T}} e^{j\omega m t}$ are orthogonal

Also, energy of $c_n \sqrt{T} \cdot \frac{1}{\sqrt{T}} e^{j\omega n t} = |c_n|^2 T$.

$$\Rightarrow \|x\|^2 = \sum_{n=-N}^{+N} |c_n|^2 \cdot T.$$

II. How do we find the coefficient c_n ?

suppose we are told that $x(t) = \sum_{n=-N}^{+N} c_n e^{j\omega n t}$

consider $(x, e^{jn\omega t})$

$$= \left(\sum_{k=-N}^{+N} c_k \cdot e^{jk\omega t}, e^{jn\omega t} \right)$$

$$= \sum_{k=-N}^{+N} (c_k \cdot e^{jk\omega t}, e^{jn\omega t})$$

$$= \sum_{k=-N}^{+N} c_k \cdot (e^{jk\omega t}, e^{jn\omega t})$$

$$= \sum_{k=-N}^{+N} c_k \cdot \sqrt{T} \cdot \sqrt{T} \left(\frac{1}{\sqrt{T}} e^{jk\omega t}, \frac{1}{\sqrt{T}} e^{jn\omega t} \right)$$

$$= \sum_{k=-N}^{+N} c_k \cdot T \cdot \delta_{k,n} = c_n \cdot T$$

$$\therefore c_n = \frac{1}{T} (x, e^{jn\omega t}) = \frac{1}{T} \int_0^T x(t) \cdot e^{-jn\omega t} dt.$$

III. Best approximation.

The signals that can be written as a linear combination of

$$e^{jn\omega t}, \quad n=0, \pm 1, \dots, \pm N$$

form a vector space of dimension $2N+1$.

We will denote this vector space as W .

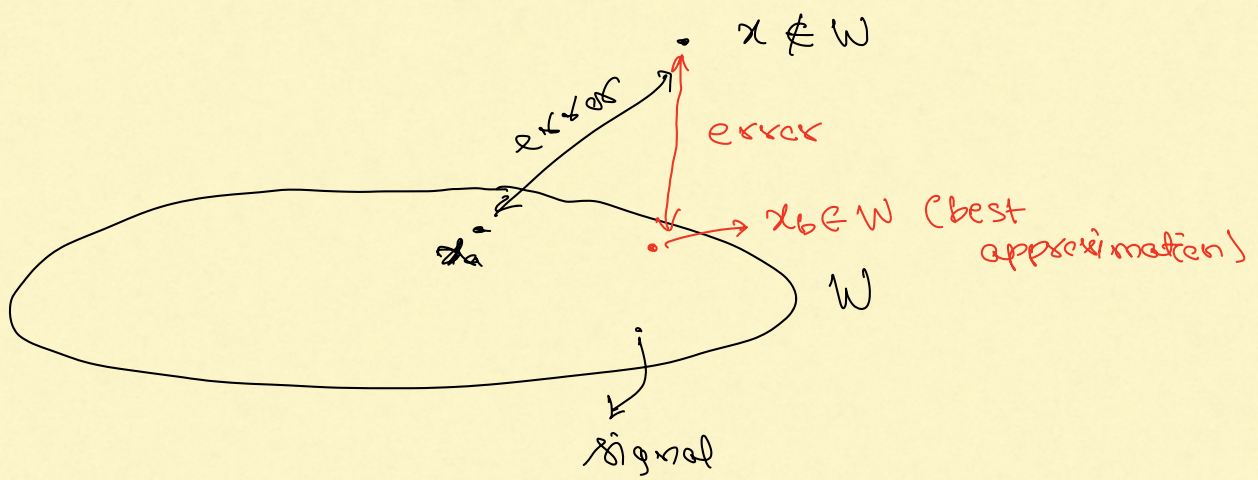
$$x, y \in W \Rightarrow x = \sum_{n=-N}^{+N} c_n \cdot e^{jn\omega t}$$

$$y = \sum_{n=-N}^{+N} b_n \cdot e^{jn\omega t}.$$

$$\alpha x + \beta y = \sum_{n=-N}^{+N} (\alpha c_n + \beta b_n) e^{jn\omega t} \in W, \text{ for any } \alpha, \beta \in \mathbb{C}$$

$\therefore W$ is indeed a vector space

(that is, W is closed under linear combinations).



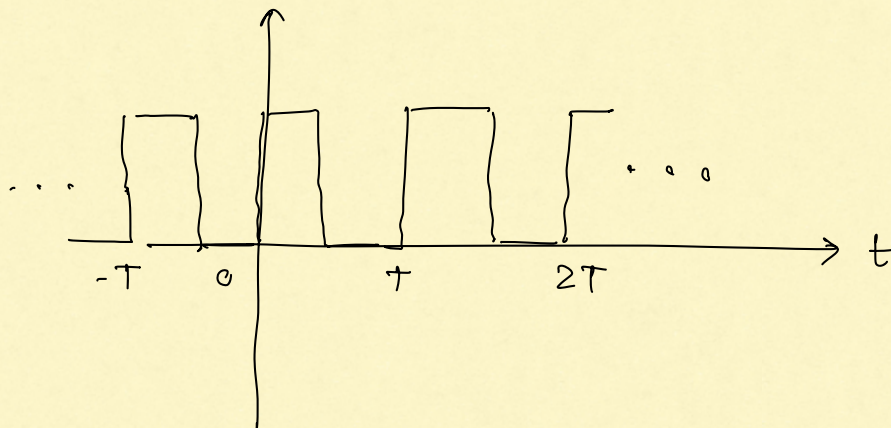
Given $x \notin W$, find $x_b \in W$ such that

$\|x - x_b\|^2$ is as small as possible.

That is, find $x_b \in W$ such that

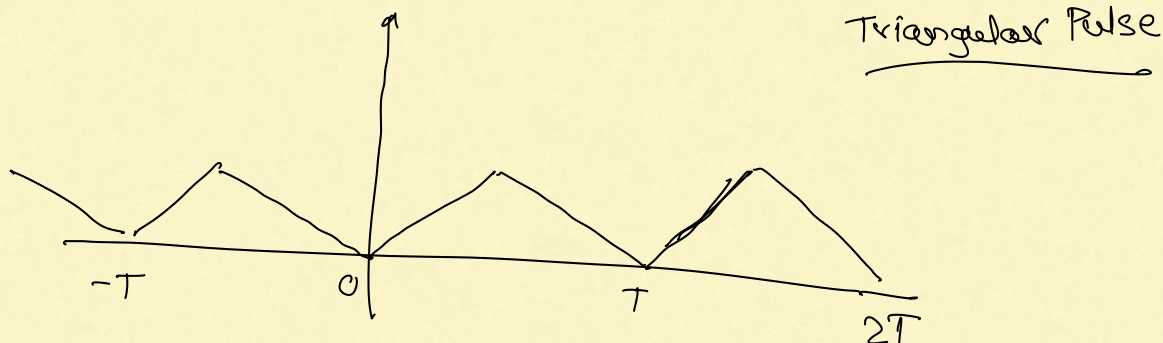
$$\|x - x_b\|^2 \leq \|x - x_a\|^2 \text{ for all } x_a \in W.$$

(1) Suppose $x(t)$ is a periodic rectangular pulse



$\hookrightarrow$ $x \notin W$ (because x is not continuous)

(2)



$x \notin W$ because it is not differentiable.